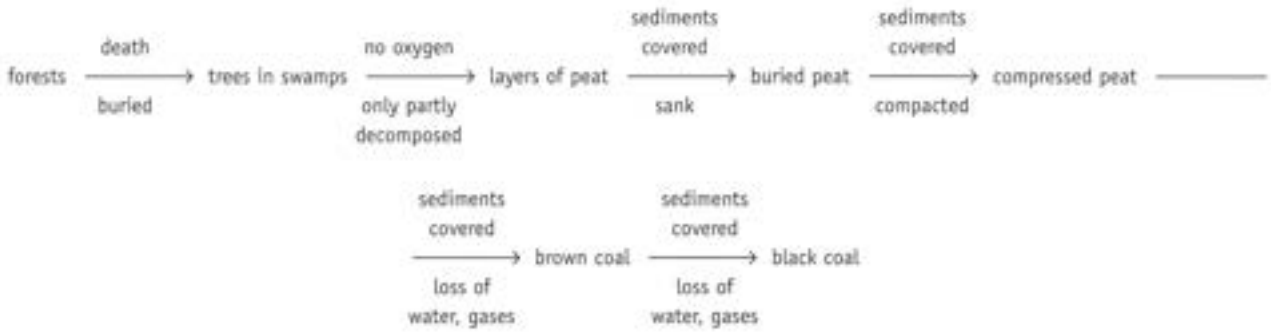
About fossil fuels

17a

Formation of coal

The flow chart below shows how coal is formed from forests over millions of years.



1 In the flow chart a substance is changed repeatedly due to some causes which affect the substance in certain ways.

a Refer to the flow chart and give two examples each.

Substance: _____ _____

Cause: _____ _____

Effect: _____ _____

b Write an explanation about coal formation. Use the scaffold below.

Title	
Summary	
Explanation sequence	
Uses of coal	

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Burning coal

When coal, which is made up of carbon, is burnt, it reacts with oxygen gas to form oxides.

2 a State where the oxygen, which is needed for the burning of coal, comes from.

_____

b Name the two possible carbon oxides, when their formulas are CO and CO_2. (*Hint:* The Greek 'mono' means one and the Greek 'di' or 'bi' means two.)

_____ _____

c Suggest which oxide is produced when there is plenty of oxygen available.

_____

d Suggest which oxide is produced when there is only some oxygen available.

_____

e CO is a poisonous gas, while CO_2 is not. If inhaled, CO will kill you, while CO_2 will only make you unconscious. Many old houses in London and Sydney have air vents in the wall. These vents are about as big as two bricks and were installed when coal fires and gas lighting were common. Explain the need for those air vents.

_____

_____

Burning of other fossil fuels

The other two groups of fossil fuels are petroleum (or crude oil) and gas. Both of these belong to the chemical group hydrocarbons.

3 a State the two elements making up petroleum and gas.

_____ _____

b When hydrocarbons burn, both their elements react with oxygen to form oxides. Name these two oxides. _____ _____

c When there is insufficient oxygen for the complete combustion of hydrocarbons, a poison and a carcinogen can be produced. The poison is made up of only two atoms and the carcinogen contains only one atom. Suggest the two unhealthy products of incomplete combustion.

_____ _____

d Petrol is also a hydrocarbon, as it is derived from petroleum. When petrol is burnt in a car engine, there can be black exhaust emitted. Suggest which substance causes the exhaust to be black. _____

e State what could be done to a car engine so that it does not perform incomplete combustion.

_____

_____

Global warming

Excessive burning of fossil fuels causes global warming.

When fossil fuels are burnt, large amounts of greenhouse gases, including carbon dioxide, are produced. These gases spread out throughout the atmosphere. When the sun's heat rays penetrate the atmosphere, they warm up the ground. The warmed ground also produces heat rays which travel away from the earth. Most of these heat rays are absorbed by the carbon dioxide, which in turn produces new heat rays. Many of these heat rays travel back to the earth causing it to become warmer.

The extra greenhouse gases in the atmosphere are causing an enhanced greenhouse effect, resulting in global warming. It is predicted that global warming will change the weather patterns, expand the oceans and melt polar ice. As a result desert areas will increase and ocean levels will rise causing flooding of low lands.

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4 **a** Look at the explanation text about global warming. State the function of each of the three paragraphs.

Paragraph 1: _____

Paragraph 2: _____

Paragraph 3: _____

b Develop a flow chart showing the events of the second paragraph. The flow chart should look like a chain.

c Develop a flow chart which shows the effects of extra greenhouse gases in the atmosphere. This flow chart should look like a hand, with arrows instead of fingers.

d Suggest how Australian society could reduce its greenhouse gas emission.

_____

_____

e State how you could reduce producing too much greenhouse gas.

_____

_____

_____

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